

Triangles

Similar figures, the Basic Proportionality Theorem and the AA, SSS and SAS criteria for similarity, following the CBSE 2026-27 syllabus.

Introduction

You have worked with congruent figures before, figures that have exactly the same shape and the same size, like two copies of the same photograph. In this chapter we look at **similar** figures. Similar figures have the same shape but not necessarily the same size. For example, all circles are similar, all squares are similar, and all equilateral triangles are similar. Two congruent figures are always similar, but two similar figures need not be congruent. This chapter focuses on similar triangles, the Basic Proportionality Theorem and the criteria used to prove two triangles similar.

Similar Figures

Two figures are called **similar** if they have the same shape. Size may differ. So two line segments are always similar, two circles are always similar, and two squares are always similar. Two rectangles, however, are not always similar, a long thin rectangle does not have the same shape as a nearly square one.

Similar Polygons

Two polygons of the same number of sides are similar if

- (i) their corresponding angles are equal, and
- (ii) their corresponding sides are in the same ratio (or proportion).

Both conditions are needed. For example, a square and a rectangle have all angles equal but their sides are not in the same ratio, so they are not similar. A square and a rhombus have sides in the same ratio but their angles are not equal, so they also are not similar.

When we write $\triangle ABC \sim \triangle DEF$, the symbol $\sim$ means "is similar to", and the order of the letters matters. $\triangle ABC \sim \triangle DEF$ means A corresponds to D, B to E, and C to F. So $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$, and $AB/DE = BC/EF = CA/FD$.

Two triangles that are congruent are certainly similar as well, since equal sides are in the ratio 1:1. But the reverse is not true, similar triangles can be of different sizes.

Basic Proportionality Theorem (Thales Theorem)

This is one of the most used results in the chapter. It connects a line drawn parallel to one side of a triangle with the ratio in which it divides the other two sides.

Basic Proportionality Theorem, Theorem 6.1 (Thales Theorem)

If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then the other two sides are divided in the same ratio.

In $\triangle ABC$, if $DE \parallel BC$ with D on AB and E on AC, then

$$AD/DB = AE/EC$$

Idea of the proof (as in NCERT). Join BE and CD, and draw $DM \perp AC$ and $EN \perp AB$. Triangles ADE and BDE share the same altitude from E to AB, and triangles ADE and CDE share the same altitude from D to AC. Using $\text{area} = \frac{1}{2} \times \text{base} \times \text{height}$:

$$\text{ar}(\triangle ADE)/\text{ar}(\triangle BDE) = AD/DB \quad \text{and} \quad \text{ar}(\triangle ADE)/\text{ar}(\triangle CDE) = AE/EC$$

But $\triangle BDE$ and $\triangle CDE$ are on the same base DE and between the same parallels DE and BC, so $\text{ar}(\triangle BDE) = \text{ar}(\triangle CDE)$. So $AD/DB = AE/EC$.

Converse of BPT, Theorem 6.2

If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side.

That is, in $\triangle ABC$, if D and E are points on AB and AC such that $AD/DB = AE/EC$, then $DE \parallel BC$.

This converse is what lets us prove that a line is parallel, by checking the ratios instead of measuring angles.

Worked Example

In $\triangle ABC$, $DE \parallel BC$. If $AD = 2$ cm, $DB = 3$ cm and $AE = 3$ cm, find EC .

By the Basic Proportionality Theorem,

$$AD/DB = AE/EC \implies 2/3 = 3/EC$$

$$EC = (3 \times 3)/2 = 9/2 = 4.5 \text{ cm}$$

Worked Example

In $\triangle ABC$, D and E are points on AB and AC such that $AD = 3$ cm, $DB = 6$ cm, $AE = 2$ cm and $EC = 4$ cm. Is $DE \parallel BC$?

$$AD/DB = 3/6 = 1/2 \quad \text{and} \quad AE/EC = 2/4 = 1/2$$

Since $AD/DB = AE/EC$, by the converse of BPT, **$DE \parallel BC$** .

Criteria for Similarity of Triangles

To check whether two polygons are similar we need both conditions, equal angles and proportional sides. For triangles, however, we do not need to check everything separately. If one condition holds, the other follows automatically. NCERT proves this through three criteria. Remember: solving a question correctly depends on naming the triangles in the right corresponding order.

AAA Similarity, Theorem 6.3

If in two triangles, corresponding angles are equal, then their corresponding sides are in the same ratio (or proportion) and so the two triangles are similar.

That is, if in $\triangle ABC$ and $\triangle DEF$, $\angle A = \angle D$, $\angle B = \angle E$ and $\angle C = \angle F$, then $AB/DE = BC/EF = CA/FD$ and $\triangle ABC \sim \triangle DEF$.

A very useful corollary follows directly from this:

AA Similarity Criterion

If two angles of one triangle are respectively equal to two angles of another triangle, then the two triangles are similar. This is called the AA criterion, and it is the same as AAA, if two angles are equal, the third angle must also be equal since the sum of angles in a triangle is 180° .

In practice, AA is the criterion you will use most often, because checking two angles is enough.

SSS Similarity, Theorem 6.4

If in two triangles, sides of one triangle are proportional to (i.e., in the same ratio as) the sides of the other triangle, then their corresponding angles are equal and so the two triangles are similar.

That is, if $AB/DE = BC/EF = CA/FD$, then $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$ and $\triangle ABC \sim \triangle DEF$.

SAS Similarity, Theorem 6.5

If one angle of a triangle is equal to one angle of the other triangle and the sides including these angles are proportional, then the two triangles are similar.

That is, if in $\triangle ABC$ and $\triangle DEF$, $AB/DE = AC/DF$ and $\angle A = \angle D$, then $\triangle ABC \sim \triangle DEF$.

Note that the angle must be the included angle, the angle formed by the two sides whose ratio you are comparing. SAS does not work with any other angle.

Worked Example

In $\triangle ABC$ and $\triangle PQR$, $\angle A = \angle P = 60^\circ$, $AB = 4$ cm, $AC = 6$ cm, $PQ = 8$ cm and $PR = 12$ cm. Are the triangles similar?

Check the sides including the equal angle:

$$AB/PQ = 4/8 = 1/2 \quad \text{and} \quad AC/PR = 6/12 = 1/2$$

So $AB/PQ = AC/PR$ and the included angle $\angle A = \angle P$. By SAS similarity, **$\triangle ABC \sim \triangle PQR$** .

Worked Example

In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. Show that $\triangle ADE \sim \triangle ABC$.

Since $DE \parallel BC$, $\angle ADE = \angle ABC$ (corresponding angles) and $\angle AED = \angle ACB$ (corresponding angles). Also $\angle A$ is common. So by AAA (or AA), **$\triangle ADE \sim \triangle ABC$** . This also shows why BPT leads to proportional sides, similarity gives $AD/AB = AE/AC = DE/BC$.

A common mistake is to assume triangles are similar just because they "look" similar. In an exam, always state the correct criterion (AA, SSS, or SAS) and write the correspondence of vertices in the correct order.

Summary

- **Similar figures** have the same shape but not necessarily the same size. Two similar polygons have equal corresponding angles and proportional corresponding sides.
- **Basic Proportionality Theorem (Thales Theorem):** if a line parallel to one side of a triangle meets the other two sides, it divides them in the same ratio. Its converse lets us prove a line is parallel by checking the ratios.
- **Criteria for similarity of triangles:** AAA (and so AA), SSS, and SAS. For SAS the equal angle must be the included angle between the proportional sides.
- **Areas of similar triangles:** $\text{ar}(\triangle ABC)/\text{ar}(\triangle PQR) = AB^2/PQ^2 = BC^2/QR^2 = CA^2/RP^2$.

Practice Questions

1. In $\triangle ABC$, $DE \parallel BC$ with $AD = 1.5$ cm, $DB = 3$ cm and $AE = 2$ cm. Find EC.

2. In $\triangle ABC$, D and E lie on AB and AC such that $AD = 4$ cm, $DB = 4.5$ cm, $AE = 8$ cm and $EC = 9$ cm. Check whether $DE \parallel BC$.

3. State and prove the Basic Proportionality Theorem (Thales Theorem).

4. In $\triangle ABC$ and $\triangle DEF$, $AB = 6$ cm, $BC = 8$ cm, $CA = 9$ cm and $DE = 9$ cm, $EF = 12$ cm, $FD = 13.5$ cm. Are the triangles similar? If so, name the similarity criterion and write the correspondence.

5. $\triangle ABC \sim \triangle PQR$ with $AB = 3$ cm and $PQ = 4.5$ cm. If $\text{ar}(\triangle ABC) = 18 \text{ cm}^2$, find $\text{ar}(\triangle PQR)$.